

Lecture 01 – chapter 24

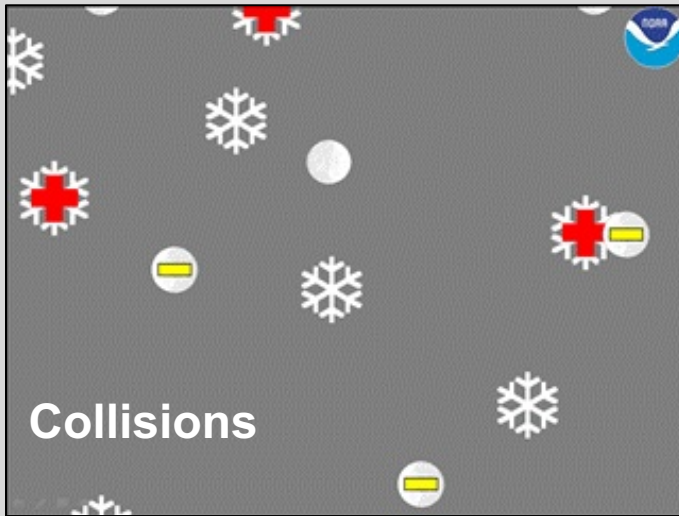
Gauss's Law

蔡尚岳

政治大學應用物理所

為了保留蔡老師的原創，以及不影響同學閱讀，
若左下角標註 CJLI，代表內容已作更動或新增。

Before Lecture



密度小、較輕的被上升
空氣往上

密度高、較重的往下，
或被上升氣流懸浮起來



密度 [編輯]

條目 討論 雙漢 臺灣正體

維基百科 - 自由的百科全書

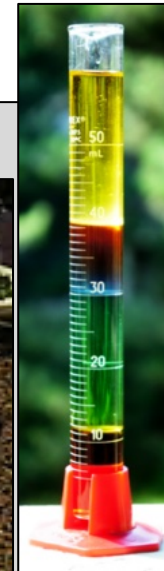
提示：此條目的主題不是紙張密度。

密度是指一物質單位體積下的質量，常用希臘字母 ρ 或是英文字母 D (Density) 表示。在數學上，密度定義為質量除以體積的商，即物體的質量與體積的比值^[1]：

$$\rho = \frac{m}{V}$$

其中 ρ 為密度， m 為質量， V 為體積。

密度	
常見符號	ρ
國際單位	kg/m^3
因次	L^{-3}M

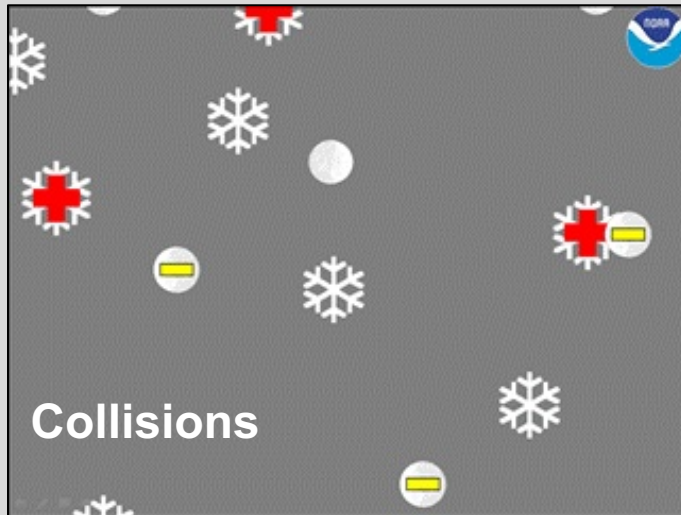


互不相溶的話，密度大的在下

<https://en.wikipedia.org/wiki/Density>
https://en.wikipedia.org/wiki/Electrostatic_discharge
<https://en.wikipedia.org/wiki/Lightning>

<https://zh.wikipedia.org/zh-tw/%E9%98%BF%E5%9F%BA%E7%B1%B3%E5%BE%B7>
<https://zh.wikipedia.org/zh-tw/%E9%BA%BB%E8%BE%A3%E7%81%AB%E9%8D%8B>

Before Lecture



密度小、較輕的被上升
空氣往上

密度高、較重的往下，
或被上升氣流懸浮起來

真假皇冠 [編輯]

主條目：[阿基米德浮體原理](#)

希倫二世國王，請金匠用純金打造了一頂純金王冠，做好了以後，吹哨者密報金匠造假摻了「白銀」在裡面，但是又不能把王冠毀壞來鑑定。阿基米德想了很久，一直沒有好方法，吃不下飯也睡不好覺。有一天，他在洗澡的時候發現，當他坐在浴盆裡時水位上升了，這使得他想到了：「上升了的水位正好應該等於王冠的體積，所以只要拿與王冠等重量的金子，放到水裡，測出它的體積，看看它的體積是否與王冠的體積相同，如果王冠體積更大，這就表示其中造了假，摻了銀。」

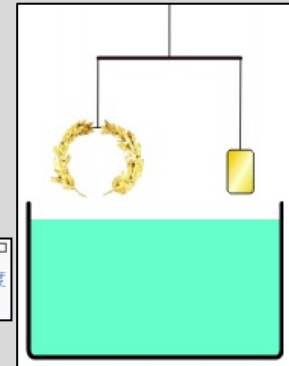
阿基米德想到這裡，不禁高興的從浴盆跳了出來，赤身裸體跑了出去，邊跑還邊喊著：「尤里卡，尤里卡！」（希臘語：[εὕρηκα](#)，意即「發現了！」）然經過證明之後，王冠中確實含有白銀，阿基米德成功的揭穿了金匠的舞弊詭計，國王對他當然是更加的信服了。

後來阿基米德將這個發現進一步總結出浮力理論，為浮體學建立了基本的定理，並寫在他的《浮體論》著作裡，也就是：物體在流體中所受的浮力，等於物體所排開的流體的重量。

阿基米德原理（或直接稱為阿基米德原理或浮力原理）是阿基米德發現的原理。該原理是說，浸在流體中的物體（全部或部分）受到豎直向上的浮力，其大小等於物體所排開流體的重量，其公式為 $F_{\text{浮力}} = G_{\text{排開流體}}$ ，公式可進一步化為 $F_{\text{浮力}} = \rho_{\text{流體}} \times g \times V_{\text{排開流體}}$ 。因為不一定作為一個理論中的「原理」，所以阿基米德原理亦稱為阿基米德定律。阿基米德浮體原理是流體力學的一個基本原理，^[1]適用於一切「全部」或「部分」浸入均勻液體的狀況。



阿基米德
前287年—前212年

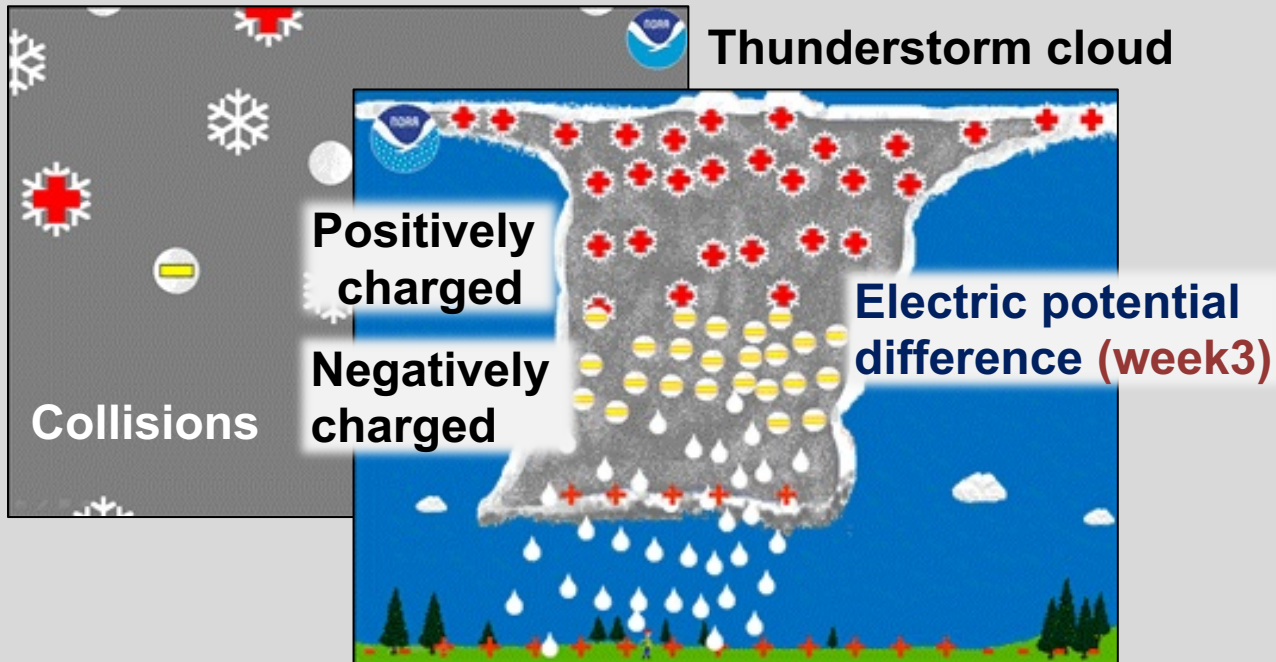


阿基米德可能使用了浮力的原理來判斷黃金王冠的密度是否小於等同質量的純金塊。

<https://en.wikipedia.org/wiki/Density>
https://en.wikipedia.org/wiki/Electrostatic_discharge
<https://en.wikipedia.org/wiki/Lightning>

<https://zh.wikipedia.org/zh-tw/%E9%98%BF%E5%9F%BA%E7%B1%B3%E5%BE%B7>

Before Lecture



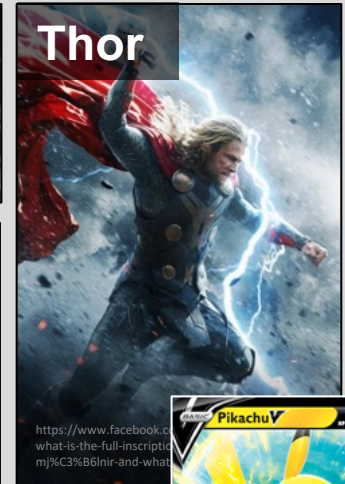
https://en.wikipedia.org/wiki/Electrostatic_discharge
<https://en.wikipedia.org/wiki/Lightning>

Previous Lecture

- About Me
 - My Research
- About “General Physics II”
 - Fascinating Electricity & Magnetism

Electric fields

- Electric charges
- Conductors and Insulators
- Coulomb’s Law
- Electric fields



This Lecture

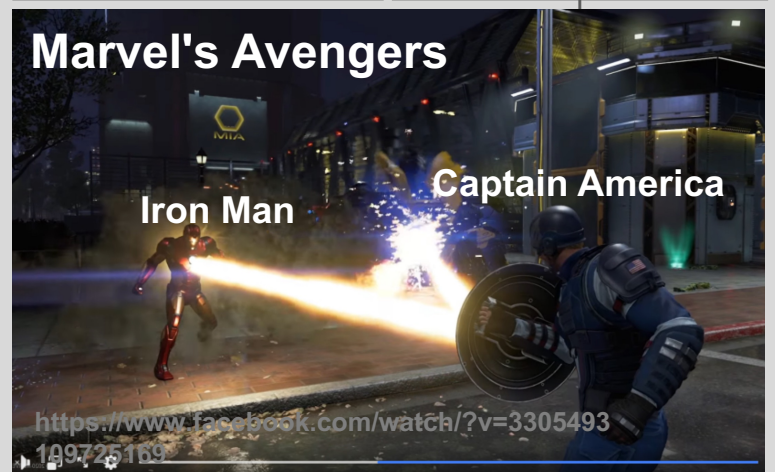
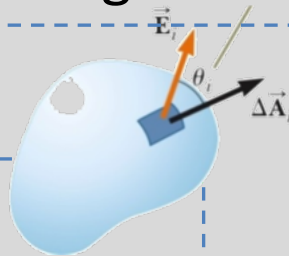
- A Story That Changed My Life
- Electric Fields of Continuous Charge Distributions

1. Superposition (sum $\Rightarrow$ integral)
 - Uniform Charged Ring & Disk

- Electric flux

2. Gauss's Law
 - Sphere, Line, Plane

Convenient for calculating the electric fields of highly symmetric charge distributions



Continuous Charge Distribution

- Charge object (任意形狀物體)
 - The total field is the sum all the small charge elements, each of which contains Δq
- For the individual charge elements

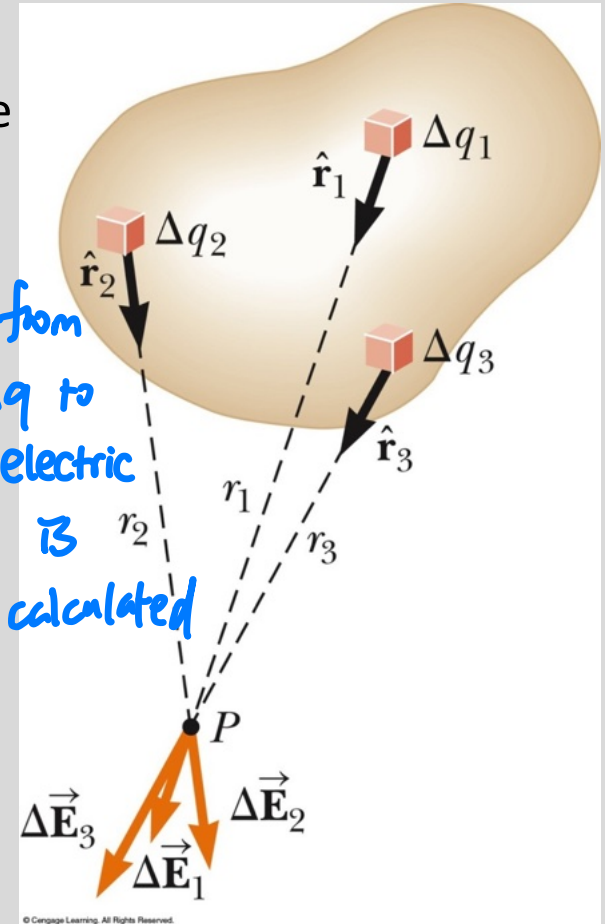
$$\Delta \vec{E} = k_e \frac{\Delta q}{r^2} \hat{r}$$

unit vector pointing from the charge element Δq to the point where the electric field is being calculated

- Because the charge distribution is continuous

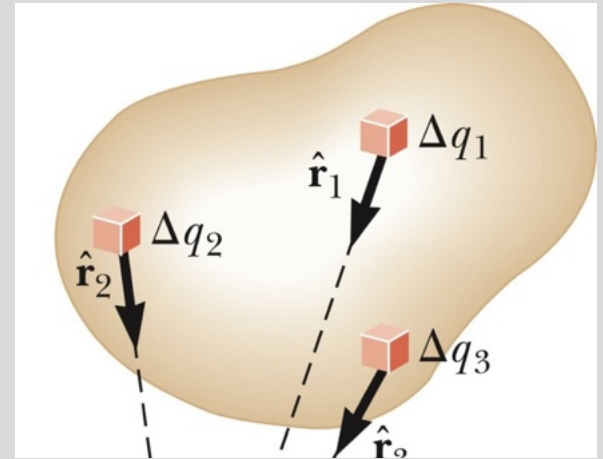
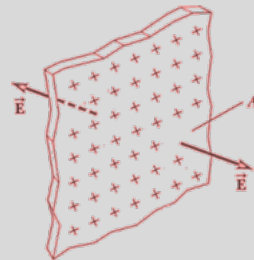
$$\vec{E} = k_e \lim_{\Delta q_i \rightarrow 0} \sum_i \frac{\Delta q_i}{r_i^2} \hat{r}_i = k_e \int \frac{dq}{r^2} \hat{r}$$

E: Electric Field



Charge Densities of Objects

- Volume charge density
 - $\rho \equiv Q / V$ with units C/m^3
- Surface charge density:
 - $\sigma \equiv Q / A$ with units C/m^2
- Linear charge density
 - $\lambda \equiv Q / \ell$ with units C/m



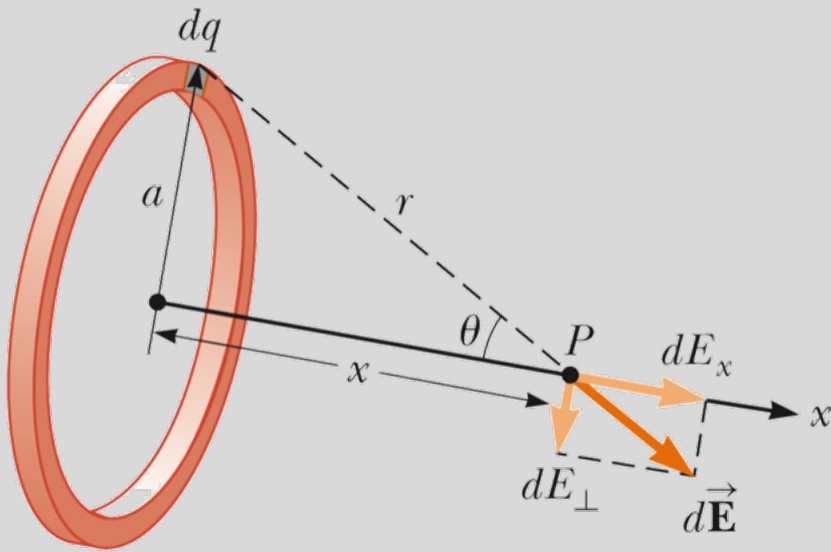
- If the charge is non-uniformly distributed over a volume, surface, or line,

- For the volume: $dq = \rho dV \longrightarrow Q = \int dq = \int \rho dV$
- For the surface: $dq = \sigma dA$
- For the length element: $dq = \lambda d\ell$

Uniform Charged Ring

- A ring of radius a carries a uniformly distributed positive total charge Q

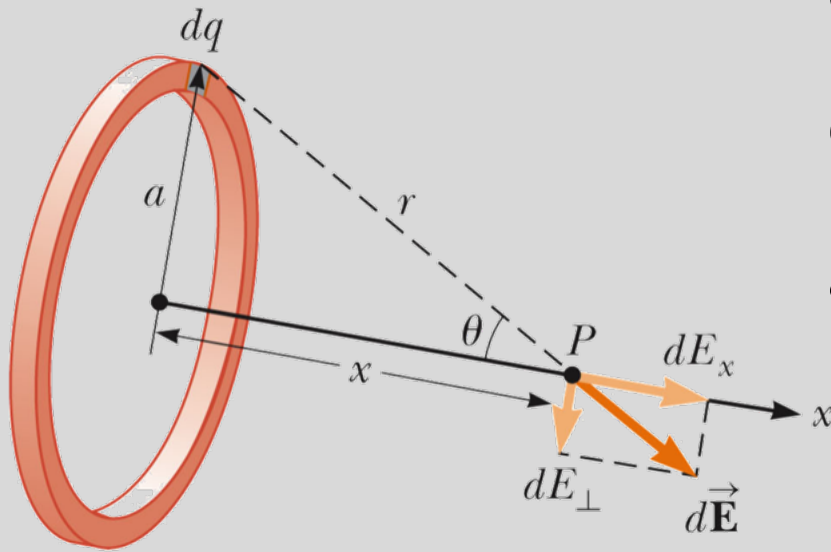
Case 1





Uniform Charged Ring

- A ring of radius a carries a uniformly distributed positive total charge Q



$$(1) \quad dE_x = k_e \frac{dq}{r^2} \cos \theta = k_e \frac{dq}{a^2 + x^2} \cos \theta$$

$$(2) \quad \cos \theta = \frac{x}{r} = \frac{x}{(a^2 + x^2)^{1/2}}$$

$$dE_x = k_e \frac{dq}{a^2 + x^2} \left[\frac{x}{(a^2 + x^2)^{1/2}} \right] = \frac{k_e x}{(a^2 + x^2)^{3/2}} dq$$

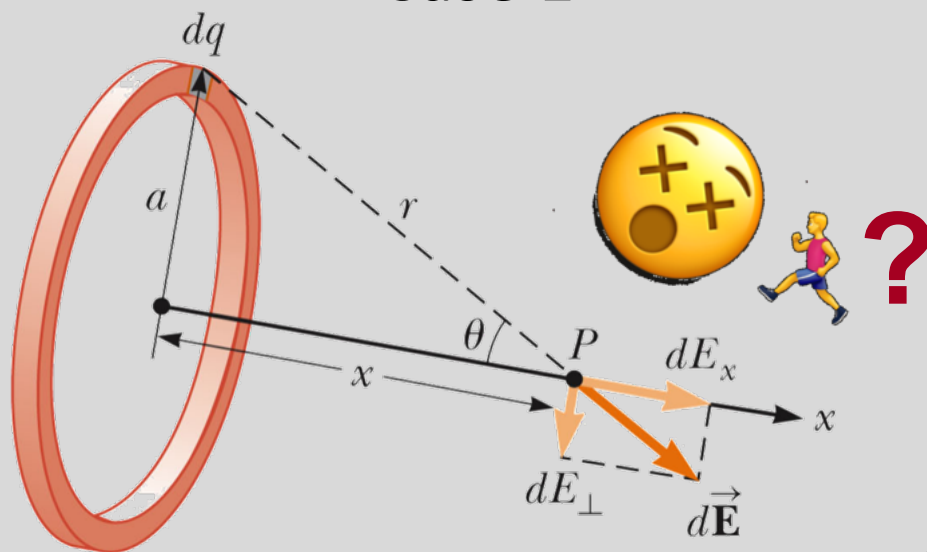
$$E_x = \int \frac{k_e x}{(a^2 + x^2)^{3/2}} dq = \frac{k_e x}{(a^2 + x^2)^{3/2}} \int dq$$

$$(3) \quad E = \frac{k_e x}{(a^2 + x^2)^{3/2}} Q$$

Uniform Charged Ring

- A ring of radius a carries a uniformly distributed positive total charge Q

Case 1



As $x = 0$, zero field

As $x \neq 0$, nonzero field

$$E_x = \int \frac{k_e x}{(a^2 + x^2)^{3/2}} dq = \frac{k_e x}{(a^2 + x^2)^{3/2}} \int dq$$

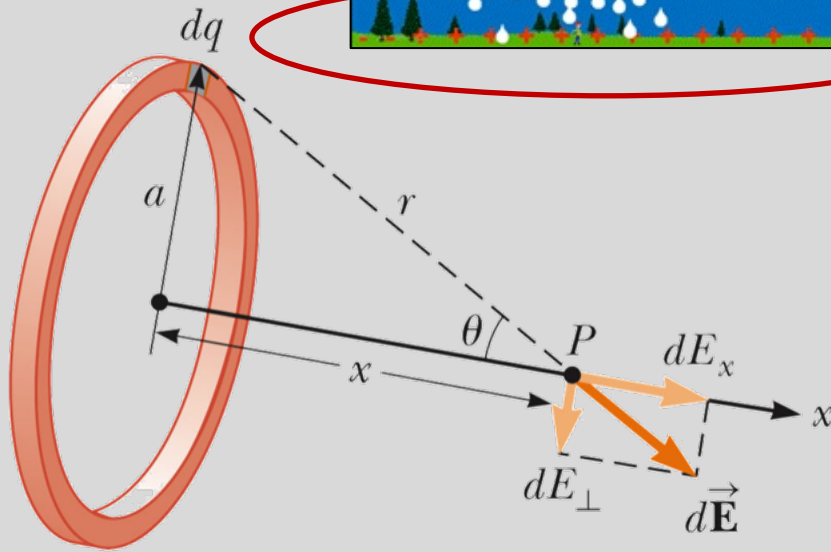
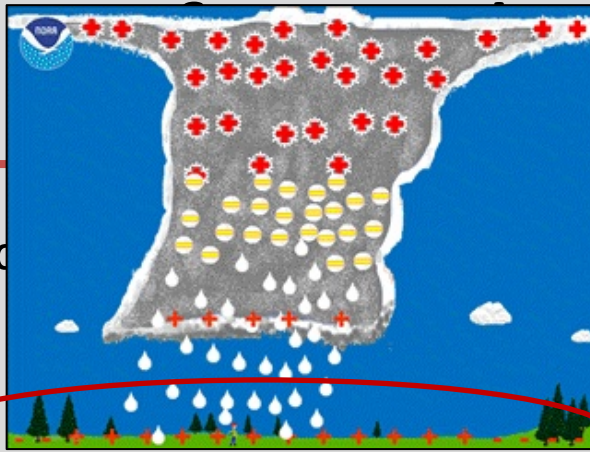
$$(3) \quad E = \frac{k_e x}{(a^2 + x^2)^{3/2}} Q$$

Thunderstorm cloud

Charged Ring

- A ring of radius a has a distributed positive charge Q

Case 2



As $x = 0$, zero field

As $x \neq 0$, nonzero field

The moongate has two rings, so it's better to run around it!

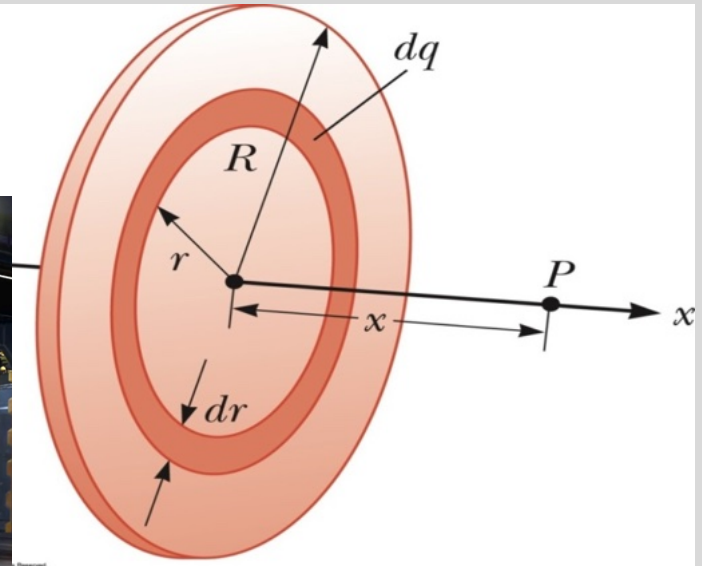
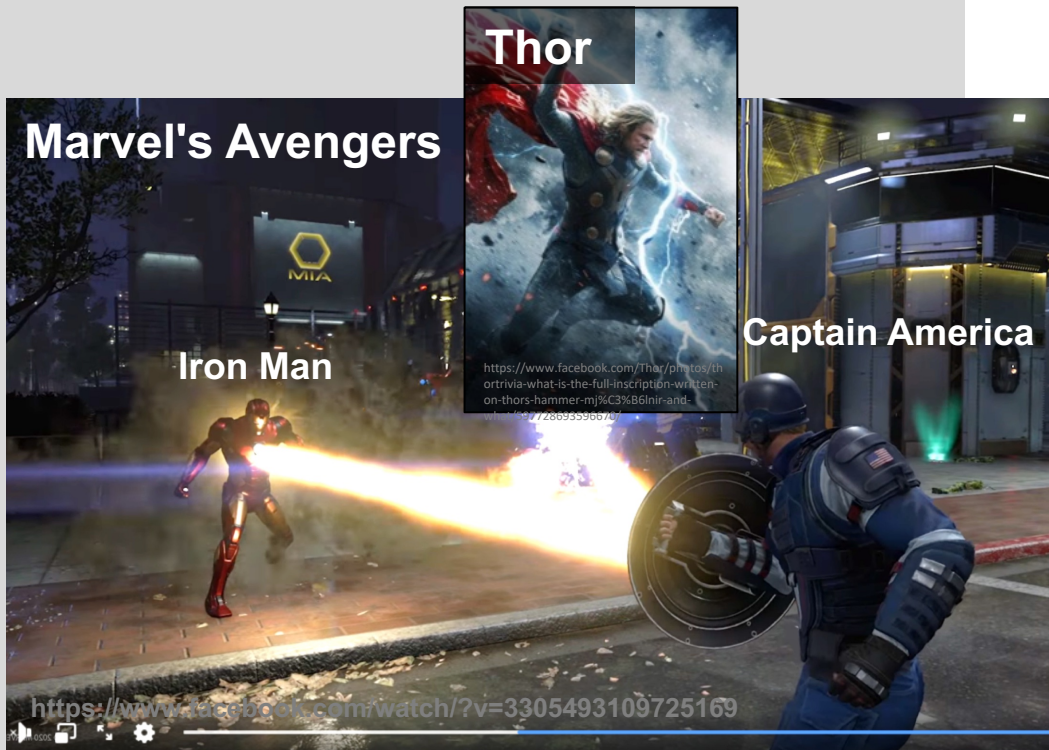
$$E_x = \int \frac{k_e dq x}{(a^2 + x^2)^{3/2}}$$

$$(3) \quad E = \frac{k_e x}{(a^2 + x^2)^{3/2}} Q$$



Uniform Charged Disk

- The disk has a radius R and a uniform charge





Uniform Charged Disk

- The disk has a radius R and a uniform charge
- Choose dq as a **ring of radius r**
- The ring has a surface area **$2\pi r dr$**

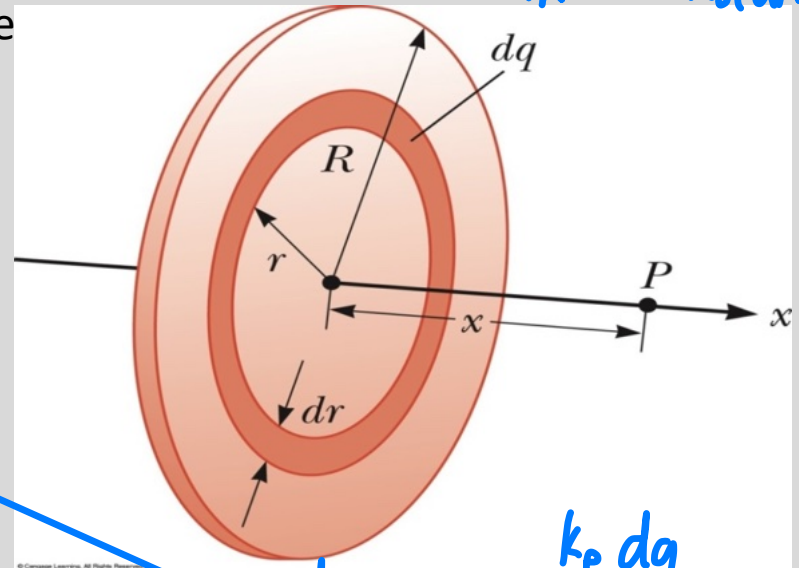
$$dq = \sigma dA = \sigma(2\pi r dr) = 2\pi\sigma r dr$$

$$dE_x = \frac{k_e x}{(r^2 + x^2)^{3/2}} (2\pi\sigma r dr)$$

$$E_x = k_e x \pi \sigma \int_0^R \frac{2r dr}{(r^2 + x^2)^{3/2}}$$

$$= k_e x \pi \sigma \int_0^R (r^2 + x^2)^{-3/2} d(r^2)$$

$$= k_e x \pi \sigma \left[\frac{(r^2 + x^2)^{-1/2}}{-1/2} \right]_0^R = 2\pi k_e \sigma \left[1 - \frac{x}{(R^2 + x^2)^{1/2}} \right]$$



σ : constant

$$dE_x = \frac{k_e dq}{r^2} \cos\theta$$

$$= \frac{k_e dq}{r^2 + x^2} \frac{x}{(r^2 + x^2)^{1/2}}$$

$$= \frac{k_e x}{(r^2 + x^2)^{3/2}} (2\pi r dr)$$

$$\begin{aligned}
 E_x &= k_e x z 6 \int_0^R \frac{2r dr}{(r^2 + x^2)^{3/2}} \\
 &= k_e x z 6 \int_{r=0}^{r=R} u^{-\frac{3}{2}} du \quad \left[\begin{array}{l} \text{Let } u = r^2 + x^2 \\ \frac{du}{dr} = 2r \\ \Rightarrow du = 2r dr \\ \Rightarrow d(r^2 + x^2) = 2r \end{array} \right. \\
 &= -2 k_e x z 6 \left[u^{-\frac{1}{2}} \right]_{r=0}^{r=R} \\
 &= -2 k_e x z 6 \left[\frac{1}{\sqrt{r^2 + x^2}} - \frac{1}{\sqrt{x^2}} \right] \\
 &= 2 k_e z 6 \left[1 - \frac{x}{\sqrt{r^2 + x^2}} \right]
 \end{aligned}$$

Electric flux (電通量)

- **Electric flux**

(phi) $\Phi_E = EA_{\perp} = EA \cos \theta = \vec{E} \cdot \vec{A}$

- **in general case**

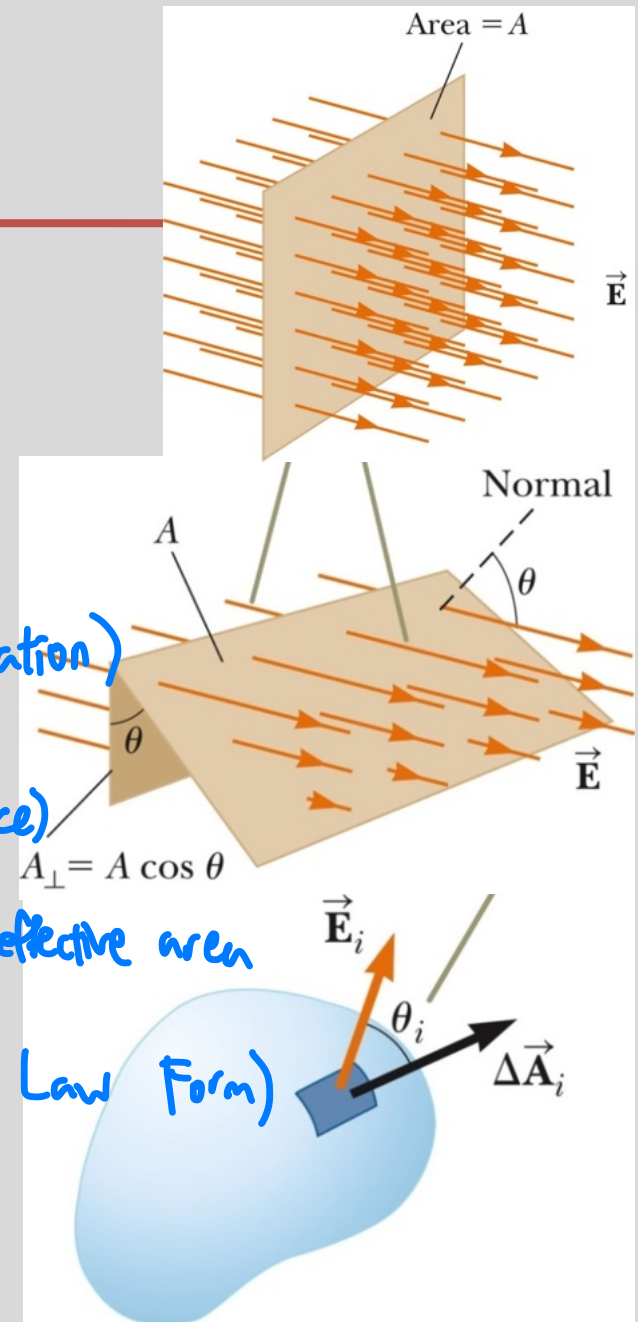
$$\Phi_E \approx \sum \vec{E}_i \cdot \Delta \vec{A}_i \quad (\text{discrete approximation})$$

$$\Phi_E \equiv \int_{\text{surface}} \vec{E} \cdot d\vec{A} \quad (\text{continuous surface})$$

- **through any close surface**

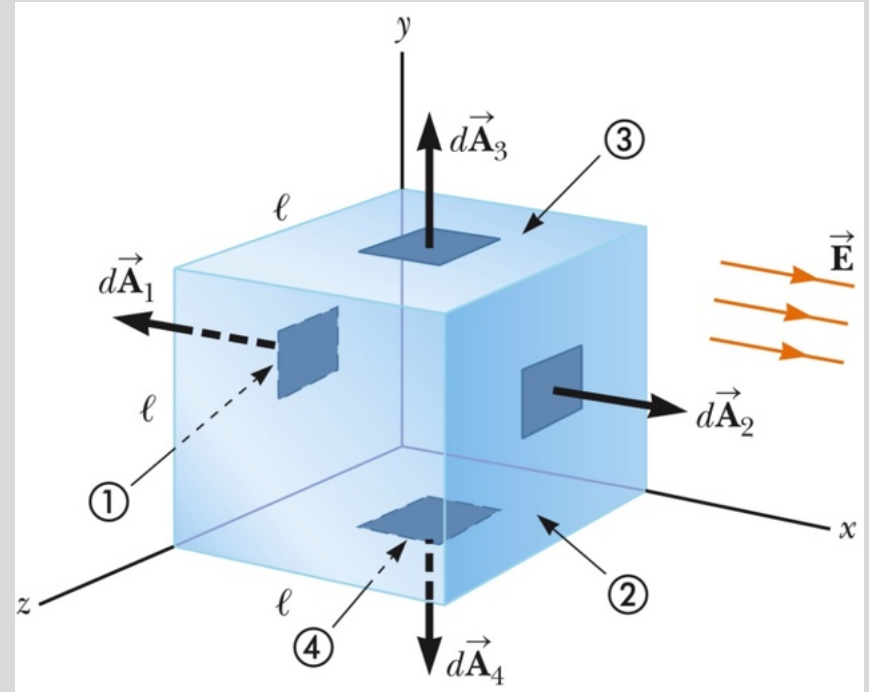
$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \int E_n dA \quad (\text{Gauss's Law Form})$$

- E_n is the component of the field perpendicular to the surface



Example 24.1 : Flux Through a Cube

- For face 1, $\phi_E = -El^2$
- For face 2, $\phi_E = El^2$
- For face 3,4, $\phi_E = 0$
- For the other sides, $\phi_E = 0$
- Therefore, $\phi_{E \text{ total}} = 0$



- ϕ_E through a closed surface that surrounds no charge is zero ???? Yes

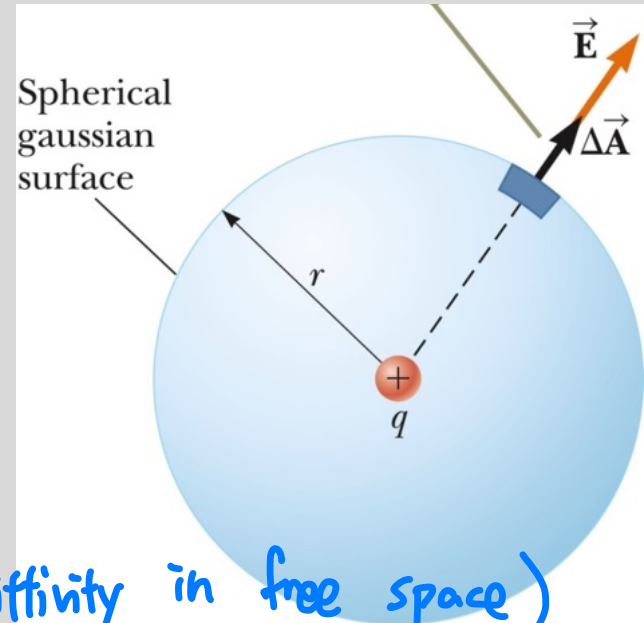
Gauss's Law – General

- The magnitude of the electric field on the surface is uniform

$$E = k_e q / r^2, A_{\text{sphere}} = 4\pi r^2$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = E \int dA$$

$$\Phi_E = 4\pi k_e q = \frac{q}{\epsilon_0} \quad (\epsilon_0 : \text{permittivity in free space})$$



- Implication :** The net electric flux through any closed surface surrounding a point charge, q , is given by q/ϵ_0
 - Independent of the shape of that closed surface (often called a *gaussian surface*)

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$k \cdot \epsilon_0 = \frac{1}{4\pi}$$

Gauss's Law – General

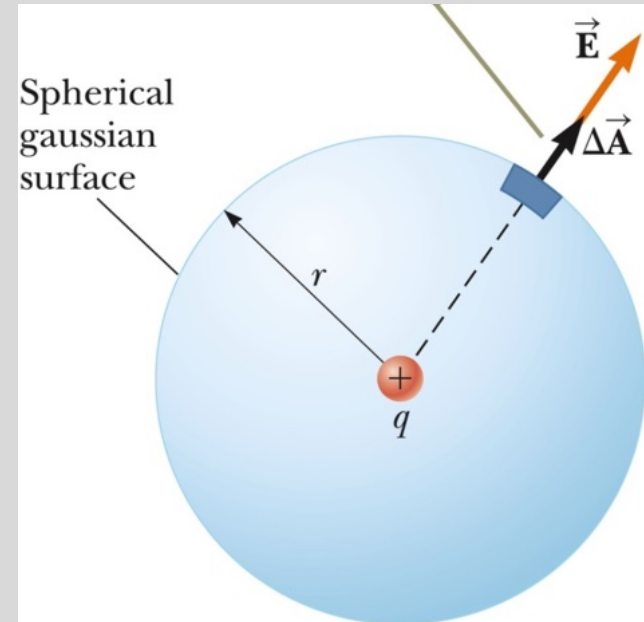
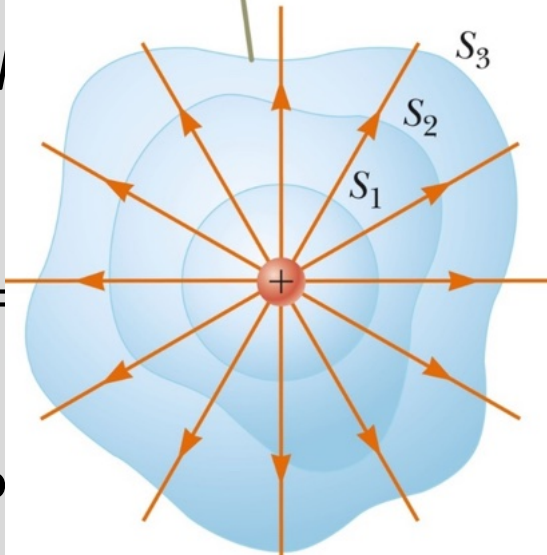
- The magnitude of the electric field is the same at all points on the surface.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0}$$

The net electric flux is the same through all surfaces.



- Implication :** The net electric flux through any closed surface surrounding a point charge, q , is given by q/ϵ_0
 - Independent of the shape of that closed surface (often called a *Gaussian surface*)

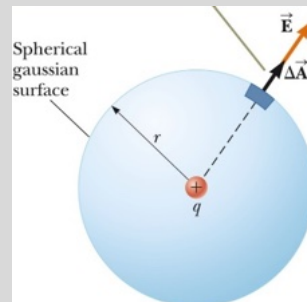
$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

Quick Quiz 24.1

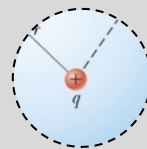
- Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere?
- (a) The flux and field both increase.
- (b) The flux and field both decrease.
- (c) The flux increases, and the field decreases.
- (d) The flux decreases, and the field increases.
- ☒ (e) The flux remains the same, and the field increases.
- (f) The flux decreases, and the field remains the same.

field:
 $kq / (r^2)$, r is halved

- Answer: (e)**



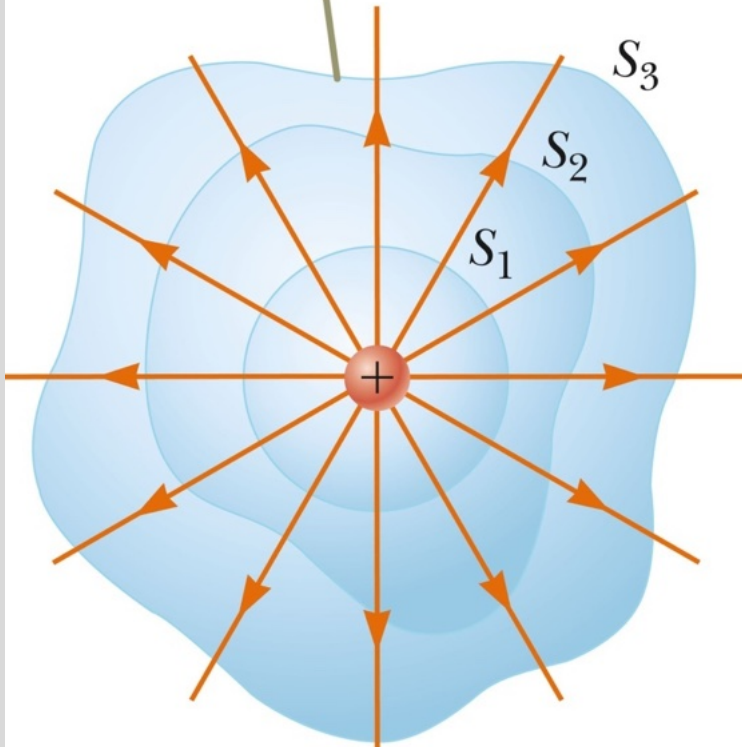
Radius is halved



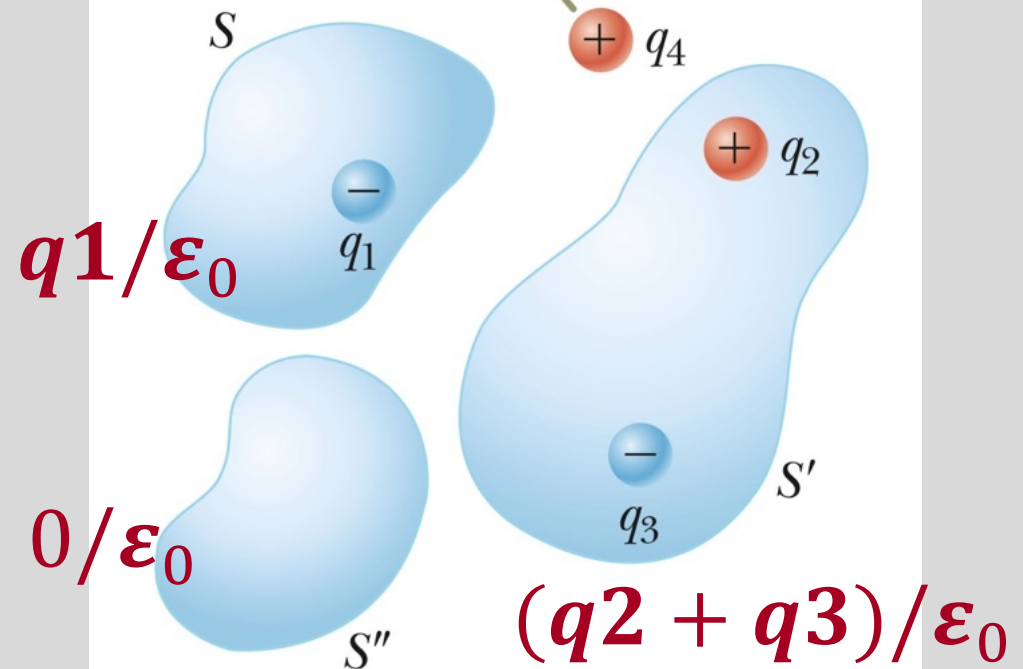
E's flux through the surface?
E's mag at the surface?

Gauss's Law

The net electric flux is the same through all surfaces.



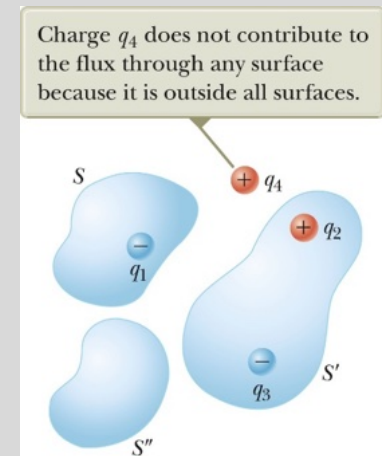
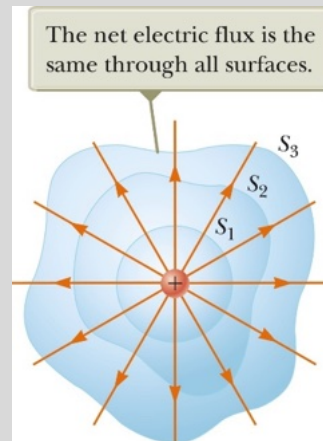
Charge q_4 does not contribute to the flux through any surface because it is outside all surfaces.



Quick Quiz 24.2

- If the net flux through a gaussian surface is zero, the following four statements *could be true*. Which of the statements *must be true*? (**multiple choices**)
there's the same amount of charges leaving the surface as the amount entering
- (a)** There are no charges inside the surface.
- (b)** The net charge inside the surface is zero.
- (c)** The electric field is zero everywhere on the surface.
- (d)** The number of electric field lines entering the surface equals the number leaving the surface.

• **Answer: (b) and (d)**



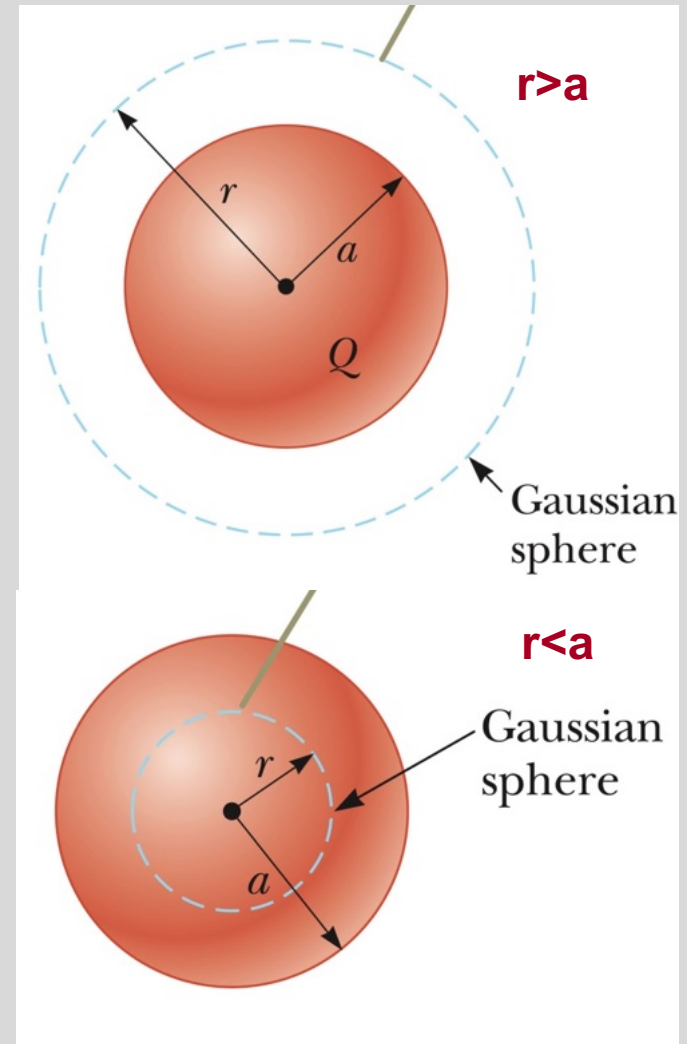
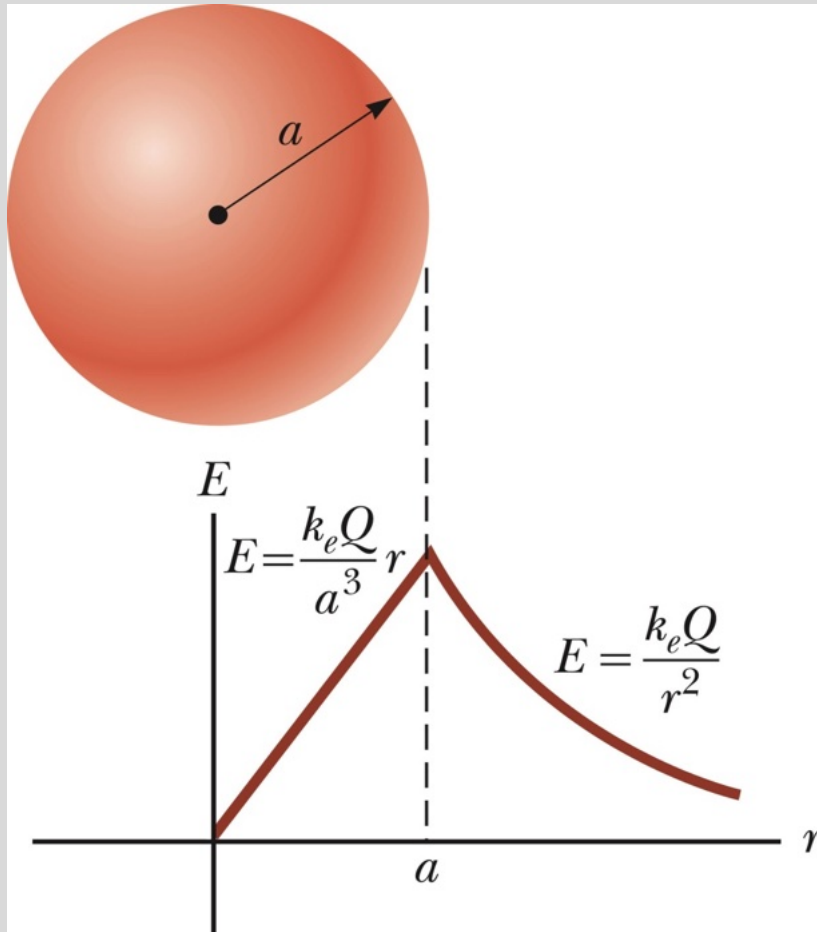
Application of Gauss's Law to Various Charge Distributions

- Useful for determining electric fields when charge distribution highly symmetric

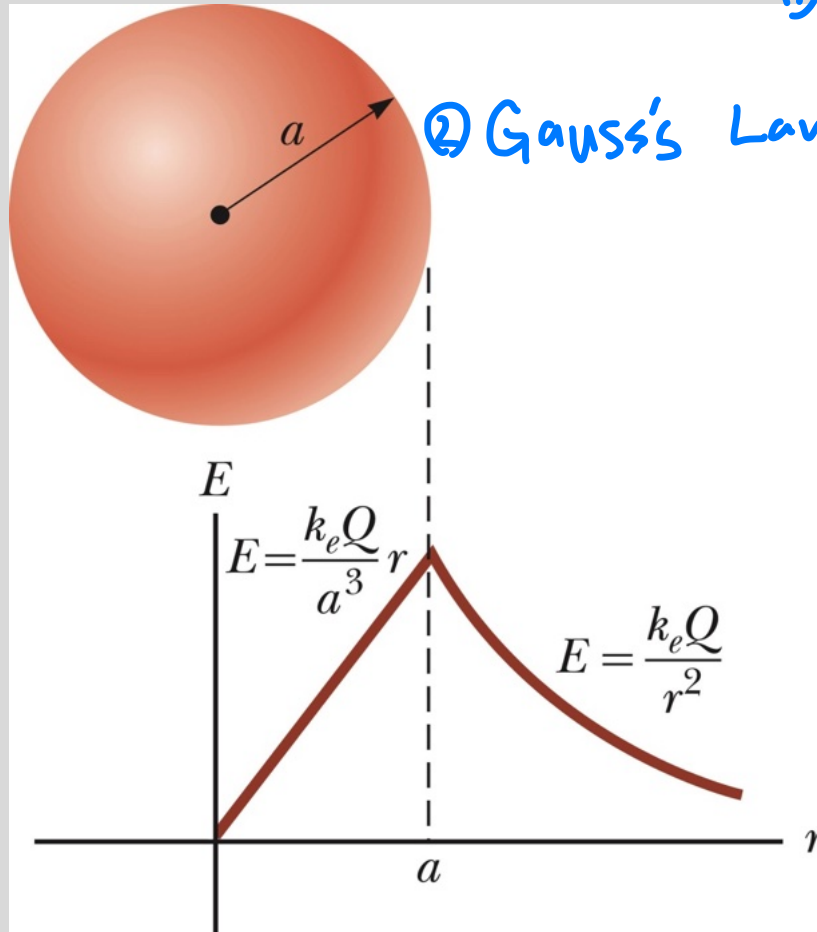
$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

- Choosing gaussian surface:
 - $\vec{E}$ constant over the portion of surface.
 - $\vec{E}$ and $d\vec{A}$ parallel: dot product = $E \, dA$
 - $\vec{E}$ and $d\vec{A}$ perpendicular: dot product = 0
 - $\vec{E} = 0$ over portion of surface

Field Due to a Spherically Symmetric Charge Distribution



Field Due to a Spherically Symmetric Charge Distribution



② Gauss's Law:

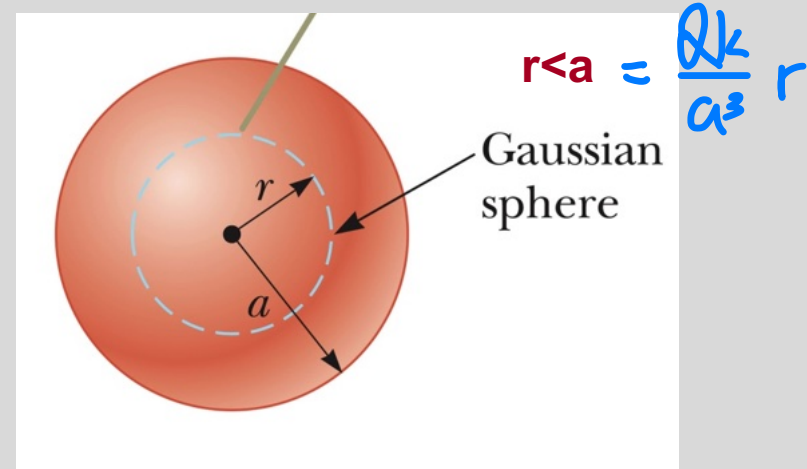
volume charge density

$$\rho = Q / \left(\frac{4}{3} \pi a^3 \right), \quad q_{in} = \rho \left(\frac{4}{3} \pi r^3 \right)$$

$$\oint E dA = E \oint dA = E (4\pi r^2) = \frac{q_{in}}{\epsilon_0}$$

$$E = \frac{q_{in}}{4\pi\epsilon_0 r^2} = \frac{\rho \left(\frac{4}{3} \pi r^3 \right)}{4\pi\epsilon_0 r^2} = \frac{\rho}{3\epsilon_0} r$$

$$E = \frac{Q / \left(\frac{4}{3} \pi a^3 \right)}{3\epsilon_0} r = \frac{Q / \left(\frac{4}{3} \pi a^3 \right)}{3(1/4\pi k)} r = \frac{Q(4\pi k)}{3 \left(\frac{4}{3} \pi a^3 \right)} r$$



Rewrite:

$$1^{\circ} \quad q_{in} = \frac{r^3}{a^3} Q$$

$$\begin{aligned} A &= 4\pi r^2 \\ V &= \frac{4}{3} \pi r^3 \end{aligned}$$

$$2^{\circ} \quad \oint E dA = \frac{q_{in}}{\epsilon_0} \quad (\text{Gauss's Law})$$

$$\Rightarrow \underline{\oint E dA = E (4\pi r^2) = \frac{r^3 Q}{a^3 \epsilon_0}}$$

$$\Rightarrow E = \frac{r Q}{4\pi a^3 \epsilon_0}$$

$$k = \frac{1}{4\pi \epsilon_0}$$

$$\Rightarrow E = \frac{k r Q}{a^3}$$

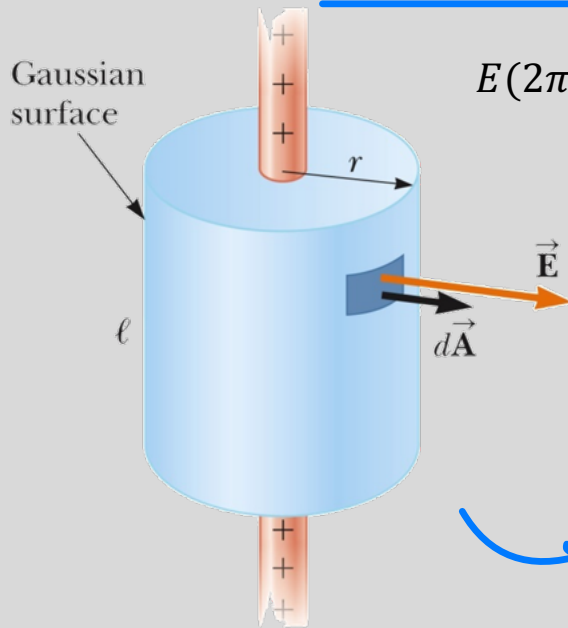
$$\Rightarrow \epsilon_0 = \frac{1}{4\pi k}$$



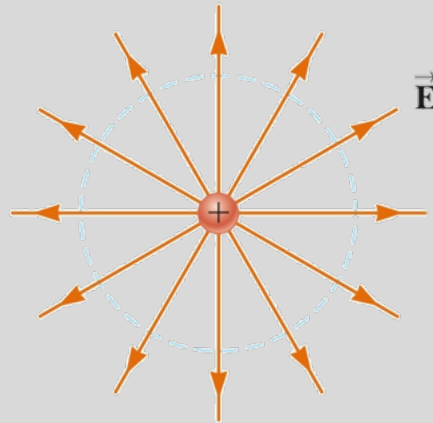
A line and a plane of charge



$$\oint E dA = E \oint dA = EA = \frac{q_{in}}{\epsilon_0} = \frac{\lambda l}{\epsilon_0}$$



$$E(2\pi r l) = \frac{\lambda l}{\epsilon_0}$$



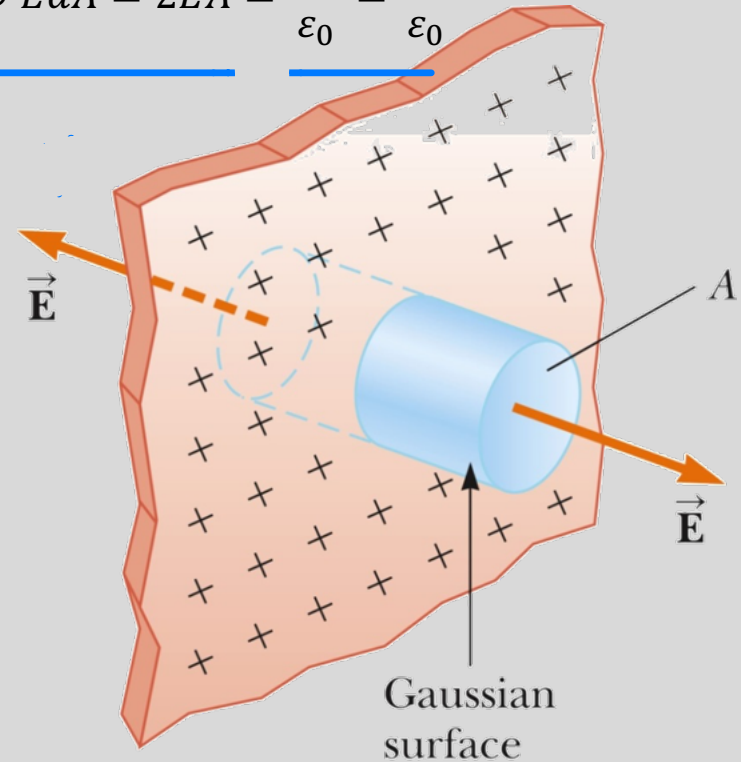
a

b

infinitely long

$$\mathbf{E} = \lambda / 2\pi\epsilon_0 \mathbf{r}$$

$$\oint E dA = 2EA = \frac{q_{in}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$



at any distance from the plane

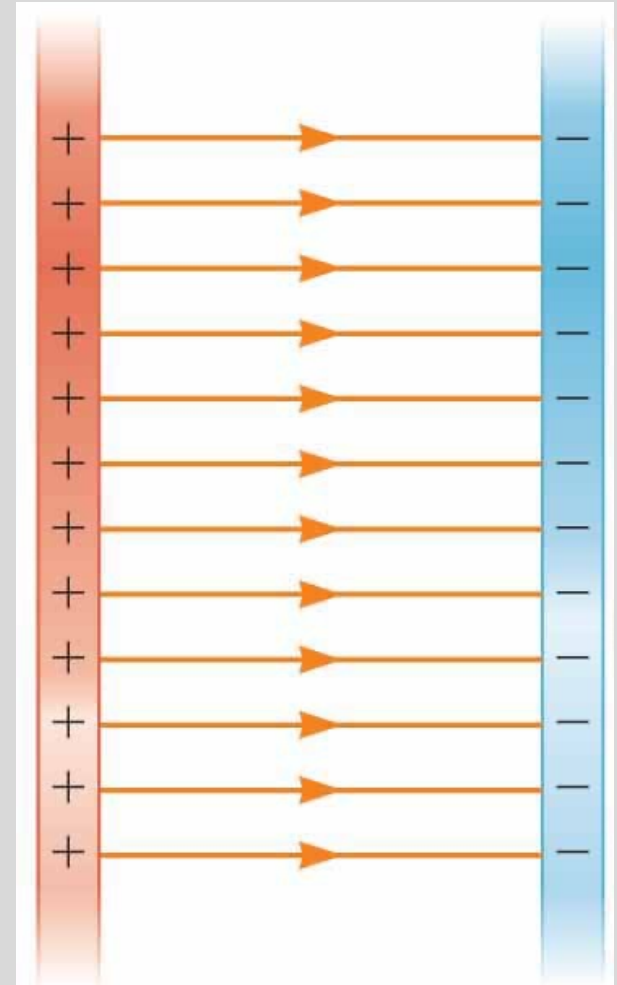
$$\mathbf{E} = \sigma / 2\epsilon_0$$



Example 23.8: A Plane of Charge

- Suppose two infinite planes of charge are parallel to each other, one positively charged and the other negatively charged. The surface charge densities of both planes are of the same magnitude. What does the electric field look like in this situation?

- $E = \sigma / \epsilon_0$: inside the planes
0 : outside the planes
(superposition)



This Lecture

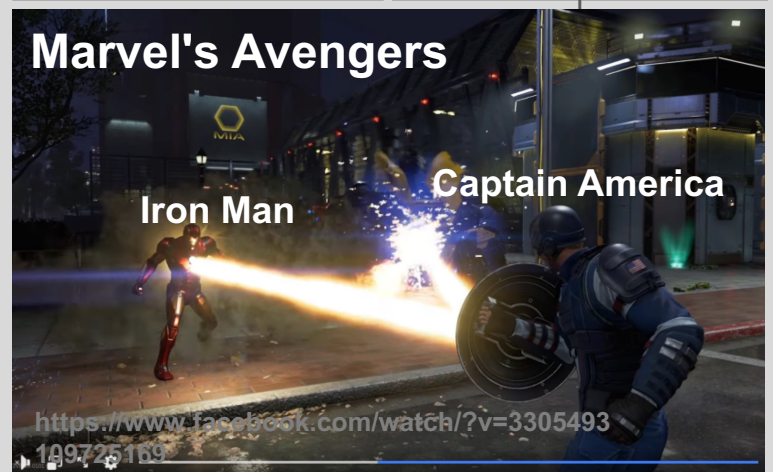
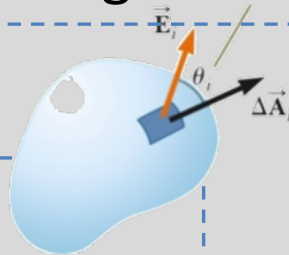
- A Story That Changed My Life
- Electric Fields of Continuous Charge Distributions

1. Superposition (sum $\Rightarrow$ integral)
 - Uniform Charged Ring & Disk

- Electric flux

2. Gauss's Law
 - Sphere, Line, Plane

Convenient for calculating the electric fields of highly symmetric charge distributions



Summary

- Charges: $q = \pm Ne$
- Coulomb's law $\vec{\mathbf{F}}_{12} = k_e \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{12}$
- Electric field, E (電場) $\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_o} = k_e \frac{q}{r^2} \hat{\mathbf{r}}$
- E of arbitrary objects $\vec{\mathbf{E}} = k_e \lim_{\Delta q_i \rightarrow 0} \sum_i \frac{\Delta q_i}{r_i^2} \hat{\mathbf{r}}_i = k_e \int \frac{dq}{r^2} \hat{\mathbf{r}}$
- Gauss' law $\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \int E_n dA \quad \Phi_E = 4\pi k_e q = \frac{q}{\epsilon_0}$